

5 (**long**) A crane is assembled from pin-jointed steel struts as shown in Fig. 5. The vertical struts in the tower and all horizontal struts are 5 m long, and the Young's modulus of the steel is 210 GPa. The crane is designed for a maximum load of 100 kN.

(a) At maximum load, calculate the bar forces in the following struts:

(i) AB, AD, AC, BD. [6]

(ii) EF, EG, FG, FH. [6]

(iii) IJ, JK, IK. [6]

(b) The crane has been designed so that every loaded strut extends or contracts by exactly 10 mm at maximum load. What is the vertical deflection of the tip of the crane (node X) at maximum load, when using this design? [12]

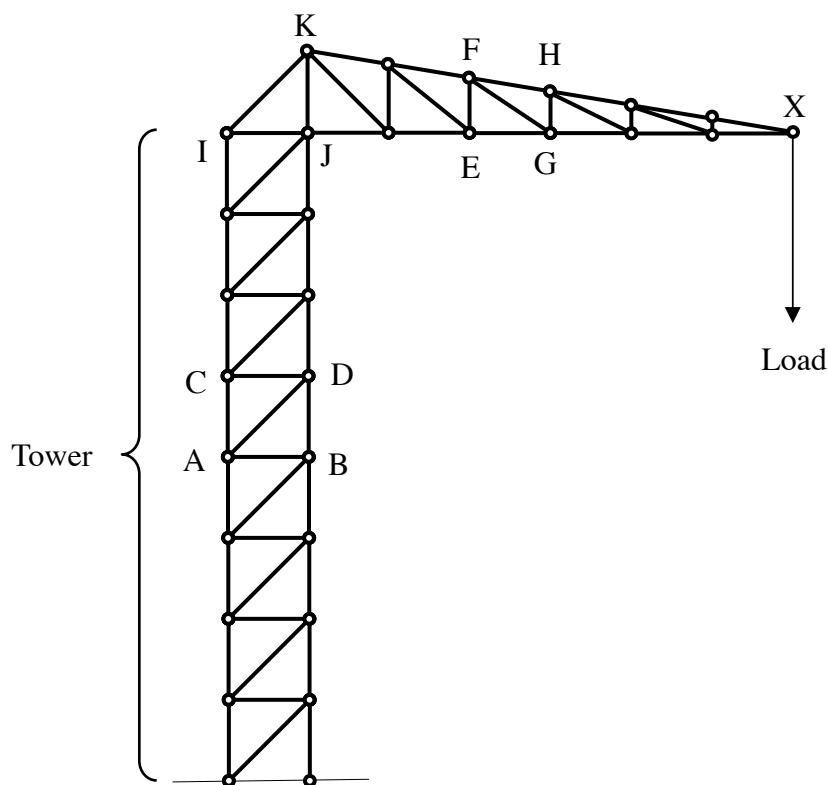


Fig. 5